

Development, simulation & implementation of precoding techniques for satellite links

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Abstract—Linear precoding has become a cornerstone of MIMO communications, as it enables spatial multiplexing gains that translate directly into higher spectral efficiency. These benefits are particularly valuable in multi-beam high-throughput satellite systems, where aggressive frequency reuse across the narrow spot beams turns the forward link into an interference-limited MU-MIMO channel. In this environment, precoding is crucial for mitigating inter-beam interference. In mobile satellite scenarios, however, the combination of long propagation delays and time-varying channels induced by user mobility introduce complications. More specific, the CSI available during the precoder computation is generally outdated with respect to the channel experienced at the time of transmission. This work investigates the performance degradation caused by this CSI mismatch and proposes a low-complexity compensation strategy. This work quantifies the impact of the CSI feedback delay on the system performance, revealing a substantial performance loss. To counter this degradation, a low-complexity prediction strategy is proposed. Numerical results show that this scheme recovers a significant fraction of the performance loss under the considered channel model.

Index Terms—MIMO, Precoding Techniques, Mobile Satellite Systems, Outdated CSI

I. INTRODUCTION

citation [1]

II. SYSTEM MODEL

III. PRECODER DESIGN

IV. OUTDATED CSI COMPENSATION

V. SIMULATION RESULTS

VI. CONCLUSION

REFERENCES

- [1] A. I. Perez-Neira, M. A. Vazquez, M. B. Shankar, S. Maleki, and S. Chatzinotas, "Signal processing for high-throughput satellites: Challenges in new interference-limited scenarios," *IEEE Signal Processing Magazine*, vol. 36, no. 4, pp. 112–131, 2019.